

BetterPresser — Technical Summary & DSP Specification

Plugin: BetterPresser (VST3 & Standalone) **Developer:** Archie DSP **Framework:** JUCE 8 (C++17)

Validation Standard: Tracktion PluginVal (Strictness Level 10)

1. Executive Summary

BetterPresser is a high-performance digital dynamic range compressor developed in C++ using the JUCE framework. The system is engineered to provide transparent, low-distortion feed-forward compression with continuous second-order polynomial soft-knee transitions, switchable Peak/RMS sidechain detection, dynamic auto-makeup compensation, and zero-allocation real-time audio thread execution. The plugin includes a resolution-independent procedural vector user interface featuring an illuminated analog VU meter with second-order physics-based needle ballistics and a live scrolling compression graph.

2. DSP Mathematics & Signal Processing Architecture



2.1. Continuous Polynomial Soft-Knee Gain Computer

To eliminate discontinuous first derivatives in the static compression curve (which creates audible harmonic distortion and clicks), BetterPresser implements a quadratic polynomial transition across knee width W :

STATIC TRANSFER CHARACTERISTIC $Y_G(X_G)$ [IN DB]

- For $x_G \leq T - \frac{W}{2}$ (Linear uncompressed region):

$$y_G(x_G) = x_G$$

- For $T - \frac{W}{2} < x_G \leq T + \frac{W}{2}$ (Polynomial transition region):

$$y_G(x_G) = x_G + \frac{(\frac{1}{R} - 1) \cdot (x_G - T + \frac{W}{2})^2}{2W}$$

- For $x_G > T + \frac{W}{2}$ (Linear compressed region):

$$y_G(x_G) = T + \frac{x_G - T}{R}$$

Target Gain Reduction: $GR_{\text{target}}(x_G) = y_G(x_G) - x_G \leq 0$ dB

2.2. Dual Detection Path (Peak & RMS)**SIDCHAIN ENVELOPE DETECTORS**

- 1. Peak Mode:** $x_{\text{lin}}[n] = \max(|s_L[n]|, |s_R[n]|)$

- 2. True RMS Mode:** Running leaky integrator with $\tau_{\text{rms}} = 30$ ms ($\alpha_{\text{rms}} = \exp(-1 / (\tau_{\text{rms}} \cdot f_s))$):

$$MS[n] = \alpha_{\text{rms}} \cdot MS[n-1] + (1 - \alpha_{\text{rms}}) \cdot \frac{1}{2} (s_L^2[n] + s_R^2[n])$$

$$x_{\text{lin}}[n] = \sqrt{\max(10^{-10}, MS[n])}$$

2.3. Decibel-Domain Envelope Ballistics**ATTACK & RELEASE DISCRETE DIFFERENCE EQUATIONS**

- **Attack Phase (when $GR_{\text{target}}[n] < Env_{\text{dB}}[n-1]$):**

$$Env_{\text{dB}}[n] = \alpha_{\text{att}} \cdot Env_{\text{dB}}[n-1] + (1 - \alpha_{\text{att}}) \cdot GR_{\text{target}}[n]$$

- **Release Phase (when $GR_{\text{target}}[n] \geq Env_{\text{dB}}[n-1]$):**

$$Env_{\text{dB}}[n] = \alpha_{\text{rel}} \cdot Env_{\text{dB}}[n-1] + (1 - \alpha_{\text{rel}}) \cdot GR_{\text{target}}[n]$$

Time constants: $\alpha_{\text{att}} = \exp(-1 / (\tau_{\text{att}} \cdot f_s))$, $\alpha_{\text{rel}} = \exp(-1 / (\tau_{\text{rel}} \cdot f_s))$

2.4. Automatic Makeup Gain Formula**DYNAMIC MATHEMATICAL COMPENSATION**

$$G_{\text{auto}}(\text{dB}) = -0.55 \cdot T \cdot (1 - \frac{1}{R}) \quad (\text{for } T \leq 0 \text{ dB})$$

3. Real-Time Thread Safety & Memory Architecture

- **Zero Heap Allocation on Audio Thread:** All audio buffers, filters, and ring buffers are statically allocated in `prepareToPlay()`. `processBlock()` performs zero `new` / `delete` operations.
- **Lock-Free Atomic Metering:** Gain reduction, input peak, and output peak meters communicate with the 60 Hz GUI thread via relaxed memory order atomics (`std::atomic<float>`).
- **APVTS State Serialization:** Full parameter layout managed through `AudioProcessorValueTreeState` with XML binary stream persistence in `getStateInformation()` / `setStateInformation()`.

4. Physics-Based VU Meter & Visualizer Engine

2ND-ORDER MASS-SPRING-DAMPER NEEDLE MODEL

$$d^2\theta/dt^2 + 2\zeta\omega_n d\theta/dt + \omega_n^2 (\theta - \theta_{\text{target}}) = 0$$

ANSI C16.5 VU parameters: Damping ratio $\zeta = 0.68$, Natural frequency $\omega_n = 22.0$ rad/s (300 ms rise time, 1.2% overshoot).

GAUSSIAN-SMOOTHED WAVEFORM VISUALIZER

$$y_{\text{smooth}}[p] = 0.25 \cdot y_{\text{raw}}[p-1] + 0.50 \cdot y_{\text{raw}}[p] + 0.25 \cdot y_{\text{raw}}[p+1]$$

Subsampled at ~6.0 ms intervals (264 samples/point @ 44.1 kHz) providing a smooth, fatigue-free ~3.3s display window.

5. Verification & Testing Conformance

#	Test Description	Input Condition	Measured Output	Status
01	Unity Ratio (1:1)	-60 dB to +12 dB range	0.000000 dB	PASS
02	Hard-Knee Mathematical Precision	Thresh -20 dB, Ratio 4:1, In -8 dB	-9.000000 dB (Exact target)	PASS
03	Soft-Knee Polynomial Continuity	Knee 10 dB, 100 sub-dB points	Continuous C ¹ transition	PASS
04	Peak vs. RMS Impulse Differentiation	1-sample 0 dBFS spike	Peak: -3.0 dB / RMS: 0.0 dB	PASS
05	Attack & Release Ballistics	Burst tone + 1s silence	Att: -14.7 dB, Rel: -0.0006 dB	PASS
06	Auto-Makeup Gain Formula	Thresh -20 dB, Ratio 4:1	+8.250000 dB	PASS
07	Dry/Wet Bit Transparency	Mix = 0% on heavy compression	Max diff = 0.000000 (Bit-exact)	PASS
08	Sidechain HPF Rejection	40 Hz sine tone, HPF @ 300 Hz	Attenuated by >16.8 dB	PASS
09	APVTS State XML Serialization	15 mutated parameters	100% restored ($\Delta < 10^{-5}$)	PASS
10	Real-Time NaN/Overload Safety	$\pm 100.0f$ (+40 dBFS) @ 96 kHz	Zero NaNs / zero infinities	PASS
11	Tracktion PluginVal	Strictness Level 10 (Max)	0 Errors / 0 Crashes	PASS

6. Project Deliverables & Output Paths

- **VST3 Plugin:** `build/BetterPresser_artefacts/Release/VST3/BetterPresser.vst3`
- **Standalone Application:** `build/BetterPresser_artefacts/Release/Standalone/BetterPresser.exe`
- **Unit Test Runner:** `build/Release/BetterPresserTests.exe`
- **Automated Test Script:** `scripts/run_tests.ps1`
- **User Guide PDF:** `docs/User_Guide.pdf`
- **Technical Summary PDF:** `docs/Technical_Summary.pdf`
- **Conformance Test Report:** `TEST_REPORT.txt` • `TEST_REPORT.md`